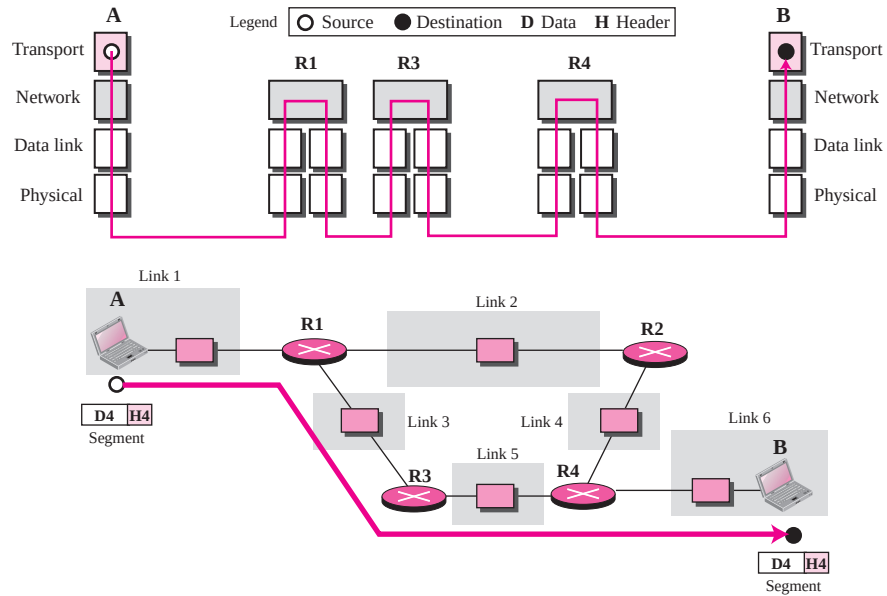


them and make a segment out of them. Figure 2.13 shows the communication at the transport layer.

Figure 2.13 *Communication at the transport layer*



Again, we should know that the two transport layers only think that they are communicating with each other using a segment; the communication is done through the physical layer and the exchange of bits.

Traditionally, the transport layer was represented in the TCP/IP suite by two protocols: **User Datagram Protocol (UDP)** and **Transmission Control Protocol (TCP)**. A new protocol called **Stream Control Transmission Protocol (SCTP)** has been introduced in the last few years.

The unit of communication at the transport layer is a segment, user datagram, or a packet, depending on the specific protocol used in this layer.

Application Layer

The application layer in TCP/IP is equivalent to the combined session, presentation, and application layers in the OSI model. The application layer allows a user to access the services of our private internet or the global Internet. Many protocols are defined at this layer to provide services such as electronic mail, file transfer, accessing the World Wide Web, and so on. We cover most of the standard protocols in later chapters. Figure 2.14 shows the communication at the application layer.